Department of Electrical and Computer Engineering
Smith Engineering Queen's University
ELEC-271, Digital Systems
Midterm
October 28, 2025

40 minutes
Total Marks: 15

Closed-book, Casio calculator allowed
Boolean algebra handout included

1. Consider the function $f(x_1, x_2, x_3) = \sum m(2,3,4,6,7)$. First, find the canonical SOP expression. Then, find the minimum SOP expression and implement it using only NAND gates. **[6]**

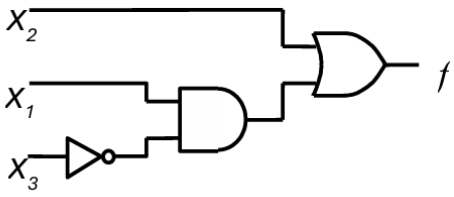
   Note: Students can use a K-map here if they want, as long as they came up with one of the reduced forms of the expression below. Marking scheme: 3 for minimum, 1.5 for SOP and 1.5 for NAND drawing

Canonical SOP expression: $f = m_2 + m_3 + m_4 + m_6 + m_7 = \bar{x}_1 x_2 \bar{x}_3 + \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 \bar{x}_3 + x_1 x_2 x_3$

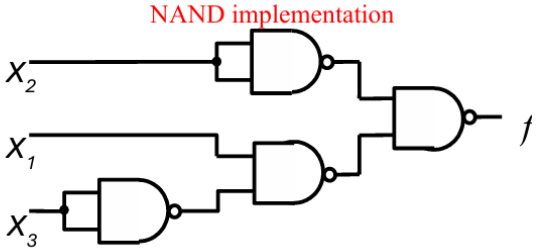
Minimum SOP: $f = \bar{x}_1 x_2(\bar{x}_3 + x_3) + x_1(\bar{x}_2 + x_2)\bar{x}_3 + x_1 x_2 x_3 = \bar{x}_1 x_2 + x_1 \bar{x}_3 + x_1 x_2 x_3 + x_1 x_2 \bar{x}_3$ Use m_6 again (7b)

$$f = \bar{x}_1 x_2 + x_1 \bar{x}_3 + x_1 x_2(x_3 + \bar{x}_3) = \bar{x}_1 x_2 + x_1 \bar{x}_3 + x_1 x_2 = (\bar{x}_1 + x_1)x_2 + x_1 \bar{x}_3 = x_2 + x_1 \bar{x}_3$$

SOP implementation



NAND implementation



2. a) Convert decimal 14.3125 to binary. **[2]**
b) Calculate the binary equivalent to 2/3 out to four places. Then convert from binary to decimal. How close is the result from 2/3. **[2]**

a)

Operation		Quotient	Remainder
14 / 2	->	7	0
7 / 2	->	3	1
3 / 2	->	1	1
1 / 2	->	0	1

Operation			binary
0.3125 x 2	->	0.625	0
0.625 x 2	->	1.25	1
0.25 x 2	->	0.5	0
0.5 x 2	->	1.0	1

1110.0101

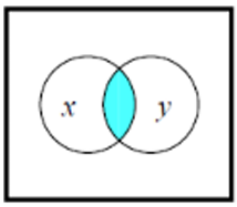
b)

Operation			binary
0.666... x 2	->	1.333...	1
0.333... x 2	->	0.666...	0
0.666... x 2	->	1.333...	1
0.333... x 2	->	0.666...	0

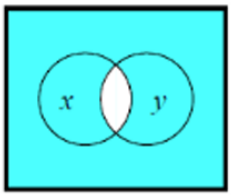
$1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 0 \times 2^{-4} = 0.625$

0.666... - 0.625 = 0.04167 approximately

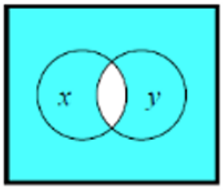
3. Use the Venn diagram to prove the DeMorgan's theorem, identity 15a. **[2]**



$x \cdot y$



$\overline{x \cdot y}$



Marking scheme: 1 mark for each side of the theorem

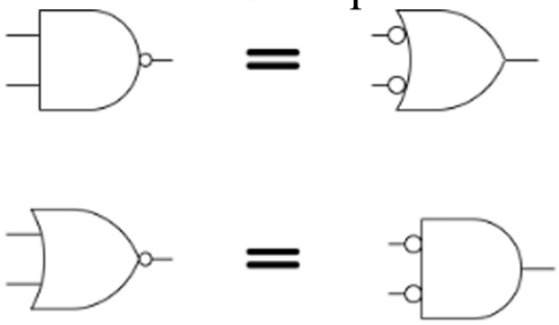
$\bar{x} + \bar{y}$

Information sheet

Conversion Table

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

NAND and NOR equivalence



Boolean Algebra

1a.	$0 \cdot 0 = 0$	1b.	$1 + 1 = 1$
2a.	$1 \cdot 1 = 1$	2b.	$0 + 0 = 0$
3a.	$0 \cdot 1 = 1 \cdot 0 = 0$	3b.	$1 + 0 = 0 + 1 = 1$
4a.	If $x = 0$ then $\overline{x} = 1$	4b.	If $x = 1$ then $\overline{x} = 0$

5a.	$x \cdot 0 = 0$	5b.	$x + 1 = 1$
6a.	$x \cdot 1 = x$	6b.	$x + 0 = x$
7a.	$x \cdot x = x$	7b.	$x + x = x$
8a.	$x \cdot \overline{x} = 0$	8b.	$x + \overline{x} = 1$
9.	$\overline{\overline{x}} = x$		

10a.	$x \cdot y = y \cdot x$	10b.	$x + y = y + x$	*Commutative*
11a.	$x \cdot (y \cdot z) = (x \cdot y) \cdot z$	11b.	$x + (y + z) = (x + y) + z$	*Associative*
12a.	$x \cdot (y + z) = x \cdot y + x \cdot z$	12b.	$x + y \cdot z = (x + y) \cdot (x + z)$	*Distributive*
13a.	$x + x \cdot y = x$	13b.	$x \cdot (x + y) = x$	*Absorption*
14a.	$x \cdot y + x \cdot \overline{y} = x$	14b.	$(x + y) \cdot (x + \overline{y}) = x$	*Combining*
15a.	$\overline{x \cdot y} = \overline{x} + \overline{y}$	15b.	$\overline{x + y} = \overline{x} \cdot \overline{y}$	*DeMorgan's Thm*
16a.	$x + \overline{x} \cdot y = x + y$	16b.	$x \cdot (\overline{x} + y) = x \cdot y$	
17a.	$x \cdot y + y \cdot z + \overline{x} \cdot z = x \cdot y + \overline{x} \cdot z$	17b.	$(x + y) \cdot (y + z) \cdot (\overline{x} + z) = (x + y) \cdot (\overline{x} + z)$	*Consensus*